



ANALYSTLAB AFRICA
Data Analytics Internship Programme
Official Weekly Assignment
Week 1: Business Analytics Case Study

Document Control

Programme	Data Analytics Internship
Week	Week 1
Project	Customer Behaviour & Business Performance Analysis
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 1 (Sunday, 11:59 PM WAT)

Business Scenario

You are a Junior Data Analyst at AnalystLab Africa Consulting. Your team has been engaged by ABC Communications Ltd to investigate customer churn and provide business insights that support customer retention strategies.

Learning Objectives

- Understand a business problem.
- Explore and analyse business datasets.
- Identify trends and patterns.
- Create professional visualisations.
- Develop business recommendations.
- Present findings professionally.

Dataset

Telco Customer Churn Dataset

<https://www.kaggle.com/datasets/blashtchar/telco-customer-churn>

Business Questions

1. What does the customer base look like?
2. Which segments have the highest churn?
3. Does contract type influence retention?
4. Does tenure affect loyalty?
5. Which services influence churn?
6. Which payment methods have higher churn?

7. What actions should management take?

Assignment Tasks

Part 1 – Business Research

Research customer churn and prepare a 1–2 page Business Understanding Report.

Part 2 – Data Loading & Inspection

Inspect rows, columns, data types, missing values, duplicates and summary statistics.

Part 3 – Business Data Analysis

Create at least 3 bar charts, 2 pie charts, 2 histograms, 1 box plot and 1 correlation heatmap with interpretations.

Part 4 – Business Insights

Provide five insights, three business risks and three opportunities.

Part 5 – Business Recommendations

Provide at least five actionable recommendations.

Deliverables

- Business Understanding Report
- Notebook (.ipynb) or Excel Workbook
- Business Presentation
- Dataset Inspection Report
- Upload to Google Drive (Anyone with the link can view)
- Submit using the official submission form

Assessment Criteria

Criteria	Weight
Business Understanding	10%
Dataset Inspection	15%
Business Data Analysis	25%
Data Visualisation	20%
Business Insights & Recommendations	15%
Professionalism	15%

Professional Development Requirement

- Update your LinkedIn profile as a Data Analytics Intern at AnalystLab Africa.
- Publish one LinkedIn post mentioning AnalystLab Africa and using #AnalystLabAfrica.
- Publish one X post tagging @analystlabafri and using #AnalystLabAfrica.
- Upload your project to GitHub with a professional README.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.

Submission Checklist

- ☐ Business Understanding Report
- ☐ Dataset Inspection Report
- ☐ Notebook/Excel Workbook
- ☐ Presentation
- ☐ GitHub Repository
- ☐ LinkedIn Post
- ☐ X Post
- ☐ Google Drive Link Submitted